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Week 4: Focal Length and Image Scale

Field Notes | Plate Scale, Field of View, and the Geometry of the Focal Plane

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Abstract

Focal length is the single geometric parameter that converts sky angle into physical image size at the telescope focal plane. This journal derives the two governing relationships from first principles: the plate scale equation s = 206.265 p/f, which gives the angular extent of one detector pixel in arcseconds, and the field-of-view equation FOV = 3438 d_s/f, which gives the angular span of the full sensor. Both equations are universal: they apply to any telescope-camera combination for which the focal length and pixel size are known, regardless of optical design, aperture, or sensor format. Focal reducers and Barlow lenses are treated as effective-focal-length multipliers within the same framework, requiring no separate derivation. To illustrate the methodology, the equations are applied to eight representative telescope-camera configurations spanning focal lengths from 490 mm to 2350 mm and image scales from 1.22 to 0.25 arcseconds per pixel. An experimental procedure for confirming any predicted plate scale by astrometric plate solving is defined with a 5 percent tolerance criterion and demonstrated on a southern-hemisphere dense star field.

Keywords: *focal length; image scale; plate scale; field of view; focal reducer; Barlow lens; telescope optics; astrophotography; pixel size; instrument trade study; southern hemisphere; angular sampling*

Nomenclature

Symbol	Definition
s	Image scale (arcseconds per pixel)
p	Pixel pitch of the camera sensor (micrometers, µm)
f	Effective focal length of the optical system (millimeters, mm)
f_{native}	Native focal length of the OTA before any optical modifier (mm); also the telescope focal length f in the visual magnification equation
f_{eff}	Effective focal length after application of a reducer or Barlow (mm)
f_e	Eyepiece focal length (millimeters, mm)
R	Focal reducer magnification factor (dimensionless; R < 1)
M_B	Barlow lens magnification factor (dimensionless; M_B > 1)
M	Visual magnification through an eyepiece (dimensionless)
FOV	Field of view along a sensor axis or eyepiece field stop (arcminutes)
d_s	Physical sensor dimension along the relevant axis (millimeters, mm)
d_{fs}	Physical diameter of the eyepiece field stop (millimeters, mm)
d_o	Object distance from the principal plane of the objective (mm)
d_i	Image distance from the principal plane of the objective (mm)
θ	Angular size of a distant object (radians unless otherwise stated)
d_{image}	Linear image size at the focal plane (millimeters, mm)
206.265	Conversion constant: 1 radian = 206265 arcseconds; dividing by 1000 converts pixel pitch from um to mm, giving 206265 / 1000 = 206.265 arcsec mm µm^-1
3438	Conversion constant: 1 radian = (180/pi) x 60 = 3437.75 arcminutes, rounded to 3438 arcmin rad^-1

1. Introduction

The Austronomic series builds the complete photon-to-pixel imaging chain one link at a time. Week 2 established light as the physical signal being collected. Week 3 established aperture as the parameter governing both photon collection rate and diffraction-limited angular resolution. Week 4 addresses the next link: the geometric mapping performed by the objective between the sky and the focal plane. Focal length is the sole governing parameter of this mapping. It determines the physical scale of the image at the focal plan, how many millimeters correspond to one degree of sky, and therefore how many arcseconds of sky each detector pixel subtends. It does not govern how many photons arrive (that depends on aperture and f-ratio, addressed in Weeks 3 and 5), nor what happens to those photons once they land on a pixel (addressed from Week 8 onward). Week 4's scope is this one geometric relationship, and it is exact.

The angular size of one pixel, expressed in arcseconds, is set by the ratio of pixel pitch to focal length. This quantity, the plate scale, governs which classes of astronomical objects are suited to a given instrument configuration. A compact globular cluster subtending 10 arcminutes in angular diameter will appear at a very different physical scale on a 490 mm system compared to a 2350 mm system, and the plate scale equation predicts that difference exactly.

The remainder of this journal derives the plate scale and field-of-view equations from geometric principles and demonstrates their application through a set of representative worked examples. The equations are instrument-independent; any reader can apply them directly to their own telescope and camera by substituting the relevant focal length and pixel size. The role of focal reducers and Barlow lenses as effective-focal-length multipliers, and the distinction between optical image scale and digital magnification, are addressed within the same universal framework.

2. Session Configuration and Experimental Objectives

The Week 4 observing session targets a Bortle 4 site in the upper Yarra Valley region, approximately 75 kilometers east of metropolitan Melbourne, Victoria. The Bortle scale is a nine-point classification of night-sky darkness, where Bortle 1 is the darkest achievable and Bortle 9 is the centre of a large city; a Bortle 4 site has dark skies with the Milky Way clearly visible and is free from significant suburban light domes. Atmospheric seeing better than 3.0 arcseconds FWHM is required; the plate scale experiment is insensitive to seeing at the level tested here, since plate solving operates on stellar centroids rather than on resolved stellar disks. A new-moon or crescent-moon window is preferred to keep sky background low, but the experiment is photometrically insensitive and can proceed under any moon phase provided the field contains adequate guide stars.

2.1 Equipment Configuration

The primary imaging configuration for this session is the Askar 71F refractor (focal length 490 mm, f/6.9) paired with the ZWO ASI585MC camera (pixel size 2.90 um, sensor 11.18 x 6.32 mm, 3856 x 2180 pixels), mounted on the EQM-35 GoTo equatorial mount. The guide subsystem is the ZWO 30F5 guide scope with the ASI220MM Mini guide camera. A supplementary run with the Skymax 150 Pro Cassegrain (focal length 1800 mm, f/12) and the same ASI585MC camera provides a high-focal-length comparison

on the same target field. These instruments are the worked examples for this journal; the methodology and equations apply equally to any telescope and camera combination.

- Primary OTA: Askar 71F, aperture 71 mm, focal length 490 mm, f/6.9.
- Camera: ZWO ASI585MC, pixel size 2.90 um, sensor 11.18 x 6.32 mm, 3856 x 2180 pixels.
- Supplementary OTA: Skymax 150 Pro, aperture 150 mm, focal length 1800 mm, f/12; manual focus only; no guiding.
- Mount: EQM-35 GoTo equatorial; polar alignment target below 5 arcminutes RMS before the imaging sequence begins.

With the instruments and constraints established, the session is organised around two specific, measurable objectives.

2.2 Session Objectives and Observing Design

The primary objective is to confirm that the plate-solve output, applied to a 60-second exposure of Omega Centauri with the primary configuration (Askar 71F + ASI585MC), agrees with the value predicted by Equation (1) to within 5 percent. The predicted plate scale is 1.22 arcseconds per pixel; the tolerance range is 1.16 to 1.28 arcseconds per pixel. A secondary objective is to record the same target field with the Skymax 150 Pro and observe the focal length dependence of plate scale directly: the predicted ratio of plate scales is 3.67, equal to the focal length ratio 1800 / 490.

The session is structured in two sequential blocks. The first block covers the primary configuration: polar alignment to below 5 arcminutes RMS, focus using a Bahtinov mask (a slotted diffraction grating placed over the objective that produces three diffraction spikes whose relative positions change precisely with focus position, allowing the point of optimal focus to be identified unambiguously) on a magnitude 2 to 4 reference star within 15 to 20 degrees of angular separation from Omega Centauri and above 45 degrees altitude, followed by a minimum 20-frame integration on Omega Centauri at 60 seconds per frame and gain 0 (the minimum electronic amplification setting, which maximises dynamic range and is appropriate here because bright stars used in the plate solve must not be saturated). The second block covers the Skymax 150 Pro: at least 20 minutes of thermal equilibration of the meniscus corrector (the thick curved lens that closes the front of the tube; its refractive index changes with temperature, so images taken before equilibration show degraded star profiles as the glass stabilises), Bahtinov mask focus on the same class of reference star, and a shorter integration sufficient for plate solving (five or more frames at 10 to 30 seconds per frame). Astrometric plate solving, the process of identifying star patterns in a captured frame and matching them against a reference catalogue to recover the frame's pointing coordinates, image scale, and orientation, is performed on uncropped raw frames; the solver-reported horizontal image scale is the measurement of record. Sigma-clipping at 3 sigma is applied to the primary integration, an iterative statistical rejection method that discards frames whose measured quality metric falls more than three standard deviations from the sample mean, removing frames degraded by cloud passages or tracking transients before final stacking [4, 5].

3. Focal Length: Geometric Foundations

A useful way to think about focal length before the mathematics: imagine the telescope objective as an arm reaching out into the sky. A longer arm concentrates the same field of sky into a smaller physical image at the focal plane, making each object appear larger but showing

less sky. A shorter arm spreads the same field across a smaller image, fitting more sky onto the sensor but representing each object at a reduced physical scale. The angular field of sky and the physical image size are linked by exactly one quantity: the focal length.

Focal length is defined as the distance from the principal plane of the optical system to the point at which parallel incoming rays converge after passing through the objective. The general relation governing image formation is the thin lens equation: $1/f = 1/d_o + 1/d_i$, where d_o is the object distance, d_i is the image distance, and f is the focal length [1, 2]. For astronomical objects, d_o is effectively infinite: $1/d_o$ approaches zero, and the equation reduces to $f = d_i$. The image forms at a distance equal to the focal length behind the objective, regardless of the object's angular size or brightness.

The physical consequence is that the angular diameter of a distant object maps to a proportional linear diameter at the focal plane. An object subtending angle θ maps to a linear image size $d_{image} = f \times \theta$. For small angles, which apply to all astronomical objects of practical interest, this simplifies to $d_{image} = f \times \theta$, with θ in radians. Doubling the focal length doubles the linear size of the image at the focal plane, while the total photon collection rate, which depends on aperture area, remains unchanged.

3.1 Visual Magnification and the Astrophotographic Equivalent

In visual observing, an eyepiece placed at the focal plane magnifies the image by the ratio $M = f_{native} / f_e$, where f_{native} is the telescope focal length and f_e is the eyepiece focal length [2]. A longer focal length telescope produces higher magnification with the same eyepiece. For astrophotography, the eyepiece is replaced by a camera sensor, and the equivalent concept is the plate scale: the angular extent, in arcseconds, of each detector pixel. The geometric relationship between focal length and image size is the same in both cases, but the plate scale formulation operates at the pixel level and does not depend on the observer.

Having established the visual analogy, the next step is to account for optical modifiers that change the effective focal length seen by the sensor.

3.2 Effective Focal Length: Reducers and Barlow Lenses

A focal reducer is a corrective lens group inserted in the converging beam before focus. It intercepts the beam and reduces the effective focal length by a factor R, where $R < 1$, giving $f_{eff} = R$ times f_{native}. A Barlow lens is a diverging element that increases the effective focal length by a factor $M_B > 1$, giving $f_{eff} = M_B$ times f_{native} [1, 3]. Both modify f_{eff} and therefore change the plate scale and field of view by the same factor. All image scale predictions must use f_{eff} rather than f_{native} for any configuration that incorporates a reducer or Barlow.

As a worked example, the instruments used in this journal include a 0.7x reducer for the Celestron EdgeHD 800 (reduced f_{eff} = 1422 mm), a 0.7x reducer for the EdgeHD 9.25 (reduced f_{eff} = 1645 mm), and a 0.8x reducer for the Askar 103APO (reduced f_{eff} = 560 mm), as well as a 2x apochromatic Barlow for the Askar 71F (Barlow-assisted f_{eff} = 980 mm). These specific configurations illustrate the calculation method; any focal reducer or Barlow can be treated identically by substituting its magnification factor R or M_B into the f_{eff} relation. The reduced and extended focal lengths in these examples feed directly into the plate scale and field-of-view calculations in Section 4.

One further distinction follows from the geometry of focal length: cropping, resampling, or digitally zooming an image after capture does not change the angular sampling encoded at the focal plane. Digital magnification increases the apparent on-screen size of objects but does not add angular resolution or modify the plate scale. Only increasing f_{eff} or reducing the pixel size p changes the true plate scale. This distinction belongs in the conceptual foundation here rather than as a consequence of any particular table: it follows directly from the geometry and is independent of which instrument or configuration is under analysis.

The geometric origin of the focal length is illustrated in Figure 1. Parallel rays from a distant source enter the objective and converge at the focal point; the focal length f is the distance from the principal plane to this convergence point.

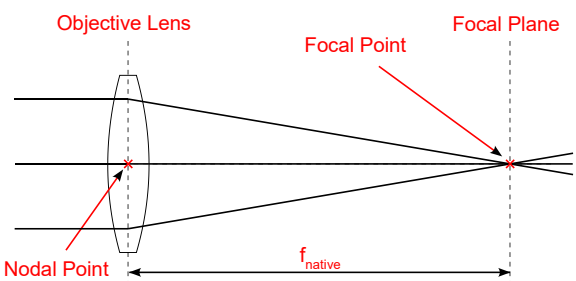

Figure 1. *Parallel rays from a distant astronomical source converging at the focal point. The focal length f is the distance from the objective principal plane to the focal point.*

Figure 1 establishes the key spatial relationship: f is not a property of the incoming light but a property of the optical system, set by the design of the objective. Every calculation in Sections 4 and 6 traces back to this single distance.

4. Image Scale and Field of View

With focal length established as the geometric scale factor for image formation, the angular size of a single detector pixel follows directly from the small-angle approximation. A pixel of physical dimension p at the focal plane of a telescope with effective focal length f subtends an angle s_{rad} = p / f at the focal point, where p and f must be in the same units. Converting to arcseconds requires multiplying by 206265 arcseconds per radian; the factor of 1000 in the denominator accounts for the unit conversion from micrometers to millimeters.

4.1 The Plate Scale Equation

The result is the plate scale equation, relating the image scale s in arcseconds per pixel to the pixel size p in micrometers and the effective focal length f in millimeters. This is Equation (1), the primary working relation for all image scale calculations across the instrument set [1, 3].

$$s = 206.265\frac{p}{f} \qquad (1)$$

s: *image scale (arcseconds per pixel)*

p: *pixel size (micrometers)*

f: *effective focal length (millimeters)*

206.265: *conversion factor (see derivation below)*

The constant 206.265 arises from two unit conversions applied in sequence. One radian contains exactly 206265 arcseconds (since 1 rad = 180 / pi degrees = 180 / pi x 3600 arcseconds = 206264.8 arcseconds, rounded to 206265). Dividing by 1000 converts the pixel size from micrometers to millimeters, matching the units of f. The combined factor is therefore 206265 / 1000 = 206.265. Equation (1) then states that the angle subtended by one pixel in arcseconds equals p[um] / f[mm] x 206.265.

Equation (1) shows that image scale is inversely proportional to effective focal length. Halving the focal length doubles the image scale; each pixel covers twice as many arcseconds of sky. As a

worked example, the Askar 71F refractor at 490 mm paired with the ZWO ASI585MC camera (2.90 um pixels) gives s = 206.265 x 2.90 / 490 = 1.22 arcseconds per pixel. A reader with a 600 mm focal length instrument and 3.76 um pixels would obtain s = 206.265 x 3.76 / 600 = 1.29 arcsec/px. The equation accepts any physically valid (f, p) pair; the calculation is identical regardless of optical design [1].

Figure 2 shows the geometric basis for Equation (1). A single pixel of width p at the focal plane subtends an angle s at the focal point; the small-angle approximation gives s_{rad} = p / f, which is the basis for the arcsecond conversion in Equation (1) [1,2].

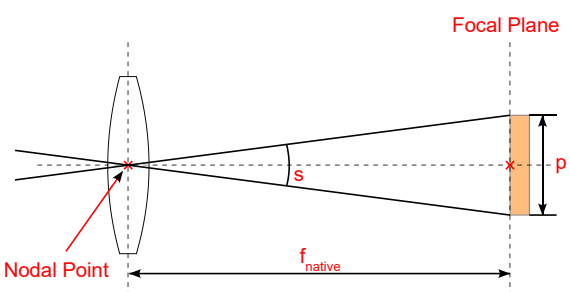

Figure 2. *The plate scale geometry. A pixel of physical dimension p at focal length f subtends an angle s = p/f radians, which yields s = 206.265 p/f arcseconds per pixel via Equation (1).*

Figure 2 makes the inverse relationship between f and s visible: as f increases, the wedge angle s narrows, and each pixel covers less sky. The same geometry applies whether f is 490 mm or 2350 mm; only the scale of the diagram changes, not its structure.

4.2 Field of View

The concept of field of view has a direct physical analogue in visual astronomy that makes the underlying geometry concrete. Every eyepiece contains a physical aperture called the field stop: a circular opening at the focal plane of the telescope that limits the cone of light reaching the eye. The angular diameter of the field stop, as seen through the eyepiece, is the true field of view. The relationship is identical to the camera case: the field stop diameter d_{fs} in millimeters divided by the telescope focal length f_{native} in millimeters gives the angular radius in radians, and multiplying by 3438 converts to arcminutes [2, 3]. A 20 mm field stop eyepiece on a 490 mm telescope gives 3438 x 20 / 490 = 140 arcminutes of true field; the same eyepiece on a 2032 mm telescope gives only 34 arcminutes. In astrophotography the camera sensor edge replaces the field stop: d_s is the sensor dimension along the relevant axis, and the governing equation is identical in form. Both cases express the same small-angle geometry: a physical dimension at the focal plane subtends an angle at the objective that scales inversely with focal length.

The full field of view of the sensor along a given axis is derived from the same geometry. If the sensor has a physical dimension d_s in millimeters along that axis, the angular extent in arcminutes follows from the small-angle approximation. The relation is Equation (2) [1, 3].

$$FOV = \frac{3438.d_s}{f} \qquad (2)$$

FOV: *field of view along the relevant sensor axis (arcminutes)*

d_s: *sensor dimension along that axis (millimeters)*

f: *effective focal length (millimeters)*

3438: *conversion factor (see derivation below)*

The constant 3438 arises from converting radians to arcminutes. One radian equals (180 / pi) degrees, and one degree equals 60 arcminutes, so 1 radian = (180 / pi) x 60 = 3437.75 arcminutes, rounded to 3438. Equation (2) therefore states that the angular span of the sensor in arcminutes equals d_s[mm] / f[mm] x

3438. The relation between Equations (1) and (2) is exact: FOV in arcminutes equals the image scale s in arcsec/px multiplied by the pixel count along that axis and divided by 60, which follows directly from the shared definitions of p, d_s, and f.

As a worked example, the ZWO ASI585MC sensor (11.18 mm x 6.32 mm) mounted on the Askar 71F at 490 mm gives a field of view of 3438 x 11.18 / 490 = 78 arcminutes (1.31 degrees) by 3438 x 6.32 / 490 = 44 arcminutes (0.74 degrees). The same sensor on a 700 mm telescope would give 55 x 31 arcminutes; on a 2000 mm telescope it would give 19 x 11 arcminutes. Equation (2) accepts any (d_s, f) pair and scales linearly: doubling f halves FOV along every sensor axis simultaneously [1].

Table 1 illustrates Equations (1) and (2) applied to three representative configurations spanning a 5.4x range of effective focal length. These configurations are drawn from instruments used in this journal; the method is identical for any telescope-camera pair. To replicate the table for a different instrument, replace f_{eff} with the telescope's focal length (or f_{eff} = R x f_{native} if a reducer is in use) and replace p and sensor dimensions with the camera datasheet values.

Table 1. *Worked example: Equations (1) and (2) applied to three configurations spanning a 3.7x range of effective focal length. The ZWO ASI585MC sensor used in this example has pixel size p = 2.90 um and physical dimensions 11.18 x 6.32 mm (3856 x 2180 pixels). Any telescope-camera pair can be evaluated identically by substituting the relevant focal length and pixel size.*

Configuration (worked example)	f_{eff} (mm)	Image Scale (arcsec/px)	FOV Width (arcmin)	FOV Height (arcmin)
Askar 71F + ASI585MC (f = 490 mm, p = 2.90 um)	490	1.22	78	44
Askar 71F + 2x Barlow + ASI585MC (f_{eff} = 980 mm)	980	0.61	39	22
Skymax 150 Pro + ASI585MC (f = 1800 mm, p = 2.90 um)	1800	0.33	21	12

Table 1 demonstrates the operational consequence directly. A factor of two increase in f_{eff} (from 490 mm to 980 mm via the 2x Barlow) halves the image scale and halves the field along every axis simultaneously. The Skymax 150 Pro at 1800 mm samples 3.7 times more finely than the Askar 71F at 490 mm when the same camera is used. These relationships hold for any instrument: the table is a template, not a specification tied to particular hardware.

Sections 3 and 4 have established the complete geometric framework: focal length determines the physical scale of the image at the focal plane, Equation (1) converts that into angular sampling per pixel, and Equation (2) converts sensor dimensions into angular field of view. Both equations are exact under the small-angle approximation and require no further assumptions. Section 5 applies this framework to define quantitative predictions that can be verified observationally.

5. Predicted Outcomes and Validation Criteria

The plate scale experiment has precisely defined predicted outcomes, computable from the instrument specifications before any data is collected. For the primary configuration used in this session (Askar 71F at 490 mm with 2.90 um pixels), the predicted image scale from Equation (1) is 1.22 arcseconds per pixel. The validation criterion is agreement between the plate-solve output and this prediction to within 5 percent; a measured value in the range 1.16 to 1.28

arcseconds per pixel constitutes confirmation. Any result outside this range is investigated using the discrepancy framework in Section 5.1.

For the supplementary high-focal-length run (Skymax 150 Pro at 1800 mm with the same 2.90 um pixels), the predicted image scale from Equation (1) is 206.265 x 2.90 / 1800 = 0.33 arcseconds per pixel. The predicted ratio of plate scales between the two configurations is 3.67, equal to the focal length ratio 1800 / 490, as Equation (1) requires when the same camera is used on both. The Omega Centauri outer halo, subtending approximately 36 arcminutes, is predicted to fill approximately half of the Askar 71F frame width (78 arcminutes wide) and nearly the full sensor width on the Skymax 150 Pro (21 arcminutes wide), providing a direct visual demonstration of the equation in the raw captured frames.

5.1 Discrepancy Analysis Framework

Discrepancies larger than 5 percent between the plate-solve result and the Equation (1) prediction are most commonly caused by one of three sources. First, an incorrect pixel size in the solver configuration: the solver must be configured with the actual pixel size of the camera being used; a mismatch shifts the reported scale in direct proportion to the error. Second, the solver reporting the pixel diagonal scale rather than the horizontal scale: the diagonal of a square pixel is larger by a factor of sqrt(2) = 1.414, which produces a reported scale approximately 41 percent higher than predicted. Third, back-focus distance error introducing field curvature that shifts the effective focal plane position; this occurs when the camera spacer stack is not set to the manufacturer's specified back-focus distance.

If none of these three sources resolves the discrepancy, the measurement is verified manually: two stars with known J2000 coordinates are identified in the frame, their angular separation is computed from the catalogue, their pixel separation is measured in the image, and the plate scale is derived as the ratio of the two. This manual result is independent of the solver configuration and serves as the ground-truth comparison. The method works for any instrument and any star field with at least two catalogued sources.

The predicted outcomes and validation criteria in this section are instrument-independent: any configuration whose focal length and pixel size are known can generate its own prediction using Equations (1) and (2), and the same 5 percent tolerance criterion and discrepancy diagnostics apply universally. Section 6 extends this to the full range of telescope-camera combinations in a systematic trade study and defines the observational procedure in detail.

6. Instrument Trade Study

The instrument trade study has two components. The analytical component applies Equations (1) and (2) to a representative set of telescope-camera combinations to quantify the image scale and field-of-view consequences of focal length selection across the full range of the instrument set. The experimental component defines a reproducible procedure for confirming any predicted plate scale by astrometric plate solving, and demonstrates the method on a specific worked example. Both components use the same governing equations and are applicable to any physically compatible telescope-camera pairing.

6.1 Analytical Component

The purpose of the analytical component is to demonstrate how Equation (1) organises any set of telescope-camera combinations into a

coherent image scale hierarchy. Image scale determines the angular extent of each pixel and therefore governs which classes of astronomical object are suited to a given configuration: a coarse image scale captures a wide field but undersamples fine angular detail, while a fine image scale resolves finer structure at the cost of a narrower field of view. The methodology is identical regardless of the specific instruments: for each combination, apply Equation (1) with the focal length and pixel size from the manufacturer datasheet.

Table 2 applies this methodology to eight representative telescope-camera configurations as worked examples. The configurations span a 4.8x range of effective focal length (490 mm to 2350 mm) and include two different pixel sizes (2.90 um and 3.76 um), covering a variety of OTA designs from short-focal-length refractors to long-focal-length Schmidt-Cassegrain telescopes with and without focal reducers. The specifications for each example are drawn from instrument datasheets. Any reader can extend this table to their own equipment by applying Equations (1) and (2) with the relevant focal length and sensor dimensions. Equations (1), (2)

The derivation chain for each row in Table 2 is identical: governing equation from Section 4 plus the instrument's focal length and pixel size equals the computed result. No intermediate steps are unique to any particular instrument.

Table 2. *Worked example: Equations (1) and (2) applied to eight representative telescope-camera configurations spanning focal lengths from 490 mm to 2350 mm. Pixel sizes: 2.90 um for ASI585MC and ASI462MC (Sony IMX585 and IMX462 sensors); 3.76 um for ASI533MC Pro (Sony IMX533 sensor). FOV long axis uses sensor long-axis dimensions: 11.18 mm (ASI585MC), 11.31 mm (ASI533MC Pro), and 5.62 mm (ASI462MC). Values are computed from manufacturer specifications; the same method applies to any telescope-camera combination.*

Configuration (worked example)	f_{eff} (mm)	Pixel (um)	Image Scale (arcsec/px)	FOV Long Axis (arcmin)
Askar 71F + ASI585MC	490	2.90	1.22	78
Askar 91F APO + ASI533MC Pro	607	3.76	1.28	64
Askar 103APO + ASI533MC Pro (native)	700	3.76	1.11	56
Askar 103APO + ASI533MC Pro (0.8x reduced)	560	3.76	1.38	69
Celestron EdgeHD 800 + ASI533MC Pro (0.7x reduced)	1422	3.76	0.55	27
Celestron EdgeHD 800 + ASI533MC Pro (native)	2032	3.76	0.38	19
Celestron EdgeHD 9.25 + ASI462MC (0.7x reduced)	1645	2.90	0.36	12
Celestron EdgeHD 9.25 + ASI462MC (native)	2350	2.90	0.25	8

Table 2 reveals two engineering insights that generalise beyond these specific examples. The first concerns the interplay between pixel size and focal length in determining image scale. The two reduced SCT configurations illustrate this: the larger telescope at 1645 mm produces a finer image scale (0.36 arcsec/px) than the smaller telescope at 1422 mm (0.55 arcsec/px), despite having a shorter reduced focal length. The reason is that the camera paired with the larger telescope has 23 percent smaller pixels (2.90 um versus 3.76 um). Applying Equation (1): both a longer focal length and a smaller pixel size independently reduce s, and here both effects act together. The ratio 0.55 / 0.36 = 1.53 equals the combined product (3.76 / 2.90) x (1645 / 1422) = 1.297 x 1.157 exactly. This result is a direct consequence of Equation (1) and applies to any pair of instruments: pixel size and focal length

are jointly determining, and neither parameter alone is sufficient to rank two configurations.

The second insight concerns the trade-off between angular resolution and sky coverage. The finest image scale in Table 2 (0.25 arcsec/px at 2350 mm native) captures only 8 x 5 arcminutes of sky per frame. That is barely large enough to frame a compact planetary nebula and far too small to contain extended objects spanning more than a degree. The widest configuration (1.22 arcsec/px at 490 mm) captures a 78 arcminute field but still undersamples fine angular structure at typical southern-hemisphere seeing. Table 2 makes this trade-off quantitative and instrument-independent: any telescope-camera combination occupies a specific (s, FOV) position determined solely by its focal length and pixel size, and the same table structure can be generated for any instrument by applying Equations (1) and (2). This connects forward to the sampling theory discussion in Week 8, where the Nyquist criterion will establish the minimum meaningful image scale for a given atmospheric seeing condition.

6.2 Experimental Component

The plate scale of any imaging system can be confirmed experimentally by astrometric plate solving a dense star field and comparing the solver-reported image scale to the value predicted by Equation (1). Astrometric plate solving is the process of identifying star patterns in a captured frame, matching them against a reference star catalogue such as the Gaia catalogue (a high-precision astrometric catalogue produced by the European Space Agency's Gaia mission, providing sub-milliarcsecond positional accuracy for over one billion stars), and computing the frame's pointing coordinates, image scale, and orientation from the match. The procedure below demonstrates the method using one specific telescope-camera pair as a worked example. To apply it to a different instrument, substitute that instrument's focal length and pixel size in step 6 to obtain the predicted value, and use any star field with sufficient Gaia catalogue coverage as the target. The tolerance criterion of 5 percent and the discrepancy diagnostics in Section 5.1 apply regardless of the instrument. Agreement is assessed against the solver-reported horizontal image scale using uncropped raw frames; cropped previews or resampled exports are excluded from the comparison.

For observers in the southern hemisphere, Omega Centauri (NGC 5139, RA 13h 26.5m, Dec -47 deg 29 arcmin, J2000) is a suitable primary target for this experiment. Its large angular diameter (approximately 36 arcminutes at the outer halo boundary) provides a rich field for plate solving, its surrounding field contains hundreds of Gaia-catalogued sources, and it transits at approximately 80 degrees altitude from the latitude of Melbourne, Victoria (-37.9 degrees south), minimising atmospheric refraction effects on astrometric positions. Omega Centauri is optimally placed from April through July; transit occurs near 23:00 AEST in late April. Observers at other latitudes should select any dense star field that transits at high altitude and contains at least a dozen Gaia-catalogued sources within the field of view of the instrument being validated.

Dark-sky sites suitable for this experiment require Bortle 3 to 4 conditions or better; the plate scale measurement itself is insensitive to sky background, but guide-star acquisition for mount tracking demands at least moderate darkness. For observers in the Melbourne, Victoria region, suitable sites within 60 to 90

minutes include the upper Yarra Valley (60 to 80 km east, Bortle 3 to 4, unobstructed southern horizon), the Macedon Ranges (approximately 60 km north-northwest, Bortle 4, good southern horizon from sites above 700 m), and the southern Mornington Peninsula (approximately 70 km south, Bortle 4, open southern horizon). Observers elsewhere should select any site with a clear horizon toward the target declination, adequate guide-star availability, and no horizon obstructions above 10 degrees elevation.

1. Set up the imaging train on the mount. Confirm that the back-focus distance is correct for the camera in use and that no unintended optical modifier (reducer, Barlow, or filter) is present in the light path unless it is part of the configuration being validated. Record the ambient temperature at setup.
2. Perform polar alignment. Target an error below 5 arcminutes RMS; larger errors will cause measurable field rotation in long exposures that can distort the plate-solve result.
3. Install the Bahtinov mask over the objective. Point the mount to a bright reference star of magnitude 2 to 4 within 15 to 20 degrees of angular separation from the target field and above 45 degrees altitude. Adjust focus until the central diffraction spike is precisely centred between the two outer spikes. Record the focuser position. Remove the Bahtinov mask before the imaging sequence.
4. Slew the mount to the selected dense star field. Confirm the target is centred in the imaging software preview.
5. Take a single exposure of sufficient duration to detect at least a dozen stars (60 seconds at gain 0 is suitable for most configurations on a Bortle 4 site). Plate-solve the frame using the imaging software plate-solve function. Record the reported image scale in arcseconds per pixel.
6. Compute the predicted image scale using Equation (1): $s = 206.265 \times p / f_{eff}$, where p is the pixel size in micrometers and f_{eff} is the effective focal length in millimeters. As a worked example, the Askar 71F at 490 mm with 2.90 um pixels gives s = 206.265 x 2.90 / 490 = 1.22 arcsec/px (tolerance range 1.16 to 1.28 arcsec/px at 5 percent). For any other instrument, substitute the relevant values. If the measured result falls outside the 5 percent tolerance, consult the discrepancy framework in Section 5.1.
7. If a focal modifier is available, install it and repeat steps 5 and 6. Compute the new predicted image scale using $f_{eff} = R \times f_{native}$ (reducer) or $f_{eff} = M_B \times f_{native}$ (Barlow), then verify that the plate-solve result shifts by the expected factor. For the worked example of the Askar 71F with a 2x Barlow, $f_{eff} = 2 \times 490 = 980$ mm, giving s = 206.265 x 2.90 / 980 = 0.61 arcsec/px (tolerance range 0.58 to 0.64 arcsec/px).
8. Record the plate-solve position angle to verify sensor orientation and confirm there is no field rotation indicative of polar alignment error exceeding the 5 arcminute target.

The expected result is the value predicted by Equation (1) for the instrument being validated, within the 5 percent tolerance range. Discrepancies are diagnosed using the framework in Section 5.1. If the solver result falls outside the tolerance, confirm by identifying two catalogued stars in the frame, computing their angular separation from J2000 coordinates, measuring their pixel separation in the image, and computing the plate scale manually as the ratio of the two.

The methodology scales to any number of telescope-camera combinations. The analytical component can be extended by adding rows to Table 2 for any instrument whose focal length and pixel size are known. The experimental component can be executed independently for each instrument using the same procedure; results from different instruments are comparable because they all reference the same governing equations and the same astrometric catalogue. When two instruments are deployed simultaneously on the same night, the ratio of their plate-solve results provides a direct observational test of the focal length ratio predicted by Equation (1), with no calibration assumptions beyond the two focal lengths and pixel sizes.

Figure 3 (optional) provides a system-level overview of the Week 4 geometry chain, linking the telescope optical train through the focal plane to the plate-solve result, and showing where reducers, Barlows, and digital post-processing each enter the chain [1].

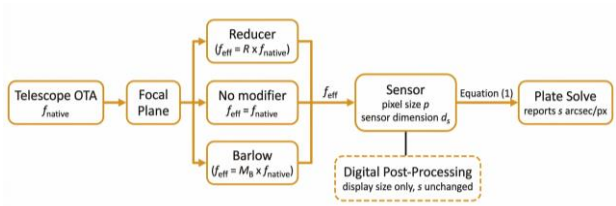

Figure 3. *The geometry chain from telescope to plate-solve result. Reducers and Barlows modify f_{eff} before the sensor; digital post-processing acts after the sensor and does not alter the recorded plate scale.*

Figure 3 makes two boundaries explicit. The boundary between the OTA and the focal plane is where reducers and Barlows act: they shift f_{eff} and therefore change both s and FOV. The boundary between the sensor and any downstream processing is where digital magnification acts: it shifts display scale only and leaves s unchanged. Keeping these two boundaries clear removes the most common source of confusion in imaging system design.

Taken together, the analytical and experimental components of this trade study provide a complete decision framework for any imaging system: compute the predicted plate scale and field of view using Equations (1) and (2), verify the prediction observationally using the procedure in Section 6.2, and diagnose any discrepancy using the framework in Section 5.1. The framework scales without modification to any telescope-camera combination for which the focal length and pixel size are known.

7. Forward Notes

The plate scale equation operates with no free parameters beyond the focal length and pixel size of the instrument. The derivation in Section 4 is exact under the small-angle approximation, which holds for all astronomical objects of practical interest. The worked examples in Tables 1 and 2 span a 7x range of focal length and two different pixel sizes, demonstrating that the equations organise any physically valid telescope-camera combination into the same coherent framework. A reader who applies Equations (1) and (2) to their own equipment will obtain predictions that can be verified to 5 percent or better using the experimental procedure in Section 6.2.

Week 5 (Focal Ratio and Exposure Efficiency) will examine the consequence of combining focal length and aperture into a single ratio. The f-ratio determines how quickly the focal plane accumulates surface brightness from extended objects, and therefore governs sub-exposure duration for a given target. The image scale analysis in Week 4 is a direct prerequisite: surface brightness per pixel depends on the solid

angle each pixel subtends, which is the square of the image scale in arcseconds per pixel.

Long-focal-length instruments such as the Skymax 150 Pro offer a high-magnification comparison that makes the focal length dependence of plate scale directly visible in a single session. Cassegrain and Maksutov-Cassegrain designs of this class require careful thermal and focus preparation: at least 20 minutes of corrector equilibration followed by Bahtinov mask focus confirmation. Planning such an instrument as the primary OTA on its own dedicated session, rather than as a supplementary run appended to a wide-field block, yields a longer integration window and a cleaner plate scale comparison at the narrow-field end of any instrument range.

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His broader objective with Austronomic is to develop a structured pathway into astronomy that moves beyond equipment-driven decision making and instead begins with conceptual clarity, technical rigor, and long-term system thinking. This work reflects his interest in building an educational framework in which astronomy can be approached not only as an observational pursuit, but also as a disciplined engineering and scientific practice.

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